

Zijian (Michael) Wang

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EDUCATION

Georgia Institute of Technology, Atlanta, GA, **GPA: 4.00/4.00**

Aug 2025 – Dec 2026

Master of Science in Statistics (ISyE)

Relevant Courses: Knowledge-based AI, Statistical Models, Experimental Design, Bayesian Statistics, Computational Analytics

Emory University, Atlanta, GA, **GPA: 3.975/4.00** with High Honor

Aug 2021 – May 2025

Bachelor of Science in Quantitative Sciences, Data Science Track, Minor in Computer Science

Relevant Courses: Machine Learning & ML Theory, Causal Inference, AI, Database, NLP, Advanced Statistical Approaches

WORK EXPERIENCE

Microsoft Research

Remote

IntentDelta: Benchmarking Agents under Evolving User Intent

Sep 2026 – Present

- Collaborating with Microsoft Research mentors to build IntentDelta, a **benchmark testing** whether **Tool-Calling LLM agents** adapt their decisions to evolving user intent in multi-turn conversations.
- Developed 400+ human-annotated intent-change trajectories with **2.6K+ conversational turns** with benchmarks including WebShop and TravelPlanner, covering five abstract update types and varied linguistic realization styles.
- Designed separate intent-level and action-level evaluations to distinguish misunderstanding from execution failure and **benchmarked 11 frontier LLMs** across 1.7K+ agent turns.

E Closet

Atlanta, GA

Data Science & AI Engineer Intern

Feb 2026 – Sep 2026

- Built and deployed a six-tier semantic search system with a self-evolving concept layer for abstract shopping queries, serving **158K production searches per month**; subsequently reduced deep-page retrieval latency by **31x (1.38s to 44ms)** and expanded reachable results per query by **7.4x (68 to 500)**.
- Designed and built a personalized **recommendation system** with vector retrieval over multi-centroid user embeddings from customer behavior, raising **Hit@10 from 10.9% to 43.0% (3.9x)** and **MRR by 4.5x** in leave-one-out evaluation.
- Developed and shipped Ask-e, a **skill-based AI customer support agent** with 14 tools spanning product search, order tracking, returns, checkout, and other shopping workflows, requiring shopper confirmation before executing actions.
- Evaluated and integrated FashionSigLIP as the **multimodal embedding backbone** for **reverse-image search**, which enabled similarity-based product retrieval from user-uploaded images and supported **~22K production requests/month**.
- Built an internal analytics platform and admin dashboard covering 50+ metrics across 8 business areas, automating code-validated daily reporting and improving revenue-forecast error (WAPE) from 0.367 to 0.339.

Buildsense AI

Remote

AI Agent Development Intern

Apr 2025 – Mar 2026

- Built PaperMem, an explainable memory system for AI agents using hybrid retrieval with pgvector and **logical graph expansion** for evidence retrieval with Neo4j; achieved 4.24/5 overall quality across 50 benchmark queries, outperforming a matched GPT-4o-mini RAG baseline by 9.9% in faithfulness and 4.9% in accuracy.
- Developed and deployed a full-stack AI research copilot with FastAPI, React, Electron, and PostgreSQL, using PaperMem to unify conversations and upload docs into memory and surface retrieved evidence through an interactive semantic graph.

Science for Georgia

Atlanta, GA

Data Science & Capstone Research Intern

Sep 2023 – May 2024

- Integrated 4+ environmental and public-health datasets for census-tract analysis using **Python** and **ArcGIS**.
- Modeled 2020–2023 water-quality trends using ARIMA/SARIMAX and benchmarked contaminants against reference thresholds, identifying 5 severe water-quality indicators above reported standards, including mercury at ~38x and E. coli at ~13x their respective thresholds.
- Developed and published an interactive ArcGIS StoryMap combining geospatial overlays, health indicators, and pollution-source analysis; identified high-risk tracts with up to 6 disadvantaged categories and 38.7% fair/poor-health prevalence to communicate findings to public audiences: [webpage](#).

RESEARCH & PROJECTS

1) “ZYM: AI Personal Trainer”

[App Link](#) | [Webpage](#)

- Built full-stack Swift iOS app and JavaScript web app, scaled to 400+ users.
- Designed a multimodal AI (skill-based agent) & CV pipeline and fine-tuned and optimized an open-source object detection model for barbell and free-weight tracking, increasing mean Average Precision (mAP@50) from 66.2% to 83.5%.

2) “Minimax NTK in Estimating Average Treatment Effect Confounded by Image Covariate”

[Honor Thesis](#)

- Designed a two-step minimax Neural Tangent Kernel estimator for causal effects under image confounding.
- Evaluated the estimator on medical imaging data with high-dimensional visual covariates.

3) “Pixel to Person: Garment Transfer with UNet Segmentation CNN”

[Link to report](#)

4) “Lyrical and Comparative Analysis of Misogyny in Rap Music with NLP”

[Presentation](#)

SKILLS

- Programming: Python, SQL, JavaScript, Swift, Java, R; AWS, Google Cloud, PyTorch, Docker, Git.
- AI/ML Systems: LLM Agents, RAG, MCP, Semantic Search, Recommendation System, NLP, ML, Benchmark Design.